

2.8.1

EE25BTECH11022 - sankeerthan

Question: If $(-4, 3)$ and $(4, 3)$ are two vertices of an equilateral triangle. Find the coordinates of the third vertex, given that the origin lies in the interior of the triangle.

solution: Let

$$\mathbf{A} = \begin{pmatrix} -4 \\ 3 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 4 \\ 3 \end{pmatrix} \text{ and } \mathbf{C} = \begin{pmatrix} x \\ y \end{pmatrix} \quad (1)$$

be the vertices of equilateral triangle.

The direction vector of the segment joining A and B is given by:

$$\mathbf{B} - \mathbf{A} = \begin{pmatrix} 4 - (-4) \\ 3 - 3 \end{pmatrix} = \begin{pmatrix} 8 \\ 0 \end{pmatrix} \quad (2)$$

The side length of equilateral triangle is

$$\|\mathbf{B} - \mathbf{A}\| = \sqrt{(\mathbf{B} - \mathbf{A})^\top (\mathbf{B} - \mathbf{A})} \quad (3)$$

$$= \sqrt{\begin{pmatrix} 8 & 0 \end{pmatrix} \begin{pmatrix} 8 \\ 0 \end{pmatrix}} \quad (4)$$

$$= \sqrt{8(8) + 0(0)} \quad (5)$$

$$= 8 \quad (6)$$

Let $\mathbf{M}$ be the midpoint of $\mathbf{A}$ and $\mathbf{B}$

$$\mathbf{M} = \frac{\mathbf{A} + \mathbf{B}}{2} \quad (7)$$

$$= \frac{\begin{pmatrix} -4 \\ 3 \end{pmatrix} + \begin{pmatrix} 4 \\ 3 \end{pmatrix}}{2} \quad (8)$$

$$= \frac{\begin{pmatrix} 0 \\ 6 \end{pmatrix}}{2} \quad (9)$$

$$= \begin{pmatrix} 0 \\ 3 \end{pmatrix} \quad (10)$$

Height of equilateral triangle

$$h = \frac{\sqrt{3}}{2} \cdot (8) = 4\sqrt{3} \quad (11)$$

The point C lies in the perpendicular bisector drawn from M. Therefore, the possible values

of C are

$$C = \begin{pmatrix} 0 \\ 3 + 4\sqrt{3} \end{pmatrix} (or) \begin{pmatrix} 0 \\ 3 - 4\sqrt{3} \end{pmatrix} \quad (12)$$

Since the origin is inside the triangle, choose the lower vertex:

$$C = \begin{pmatrix} 0 \\ 3 - 4\sqrt{3} \end{pmatrix} \quad (13)$$

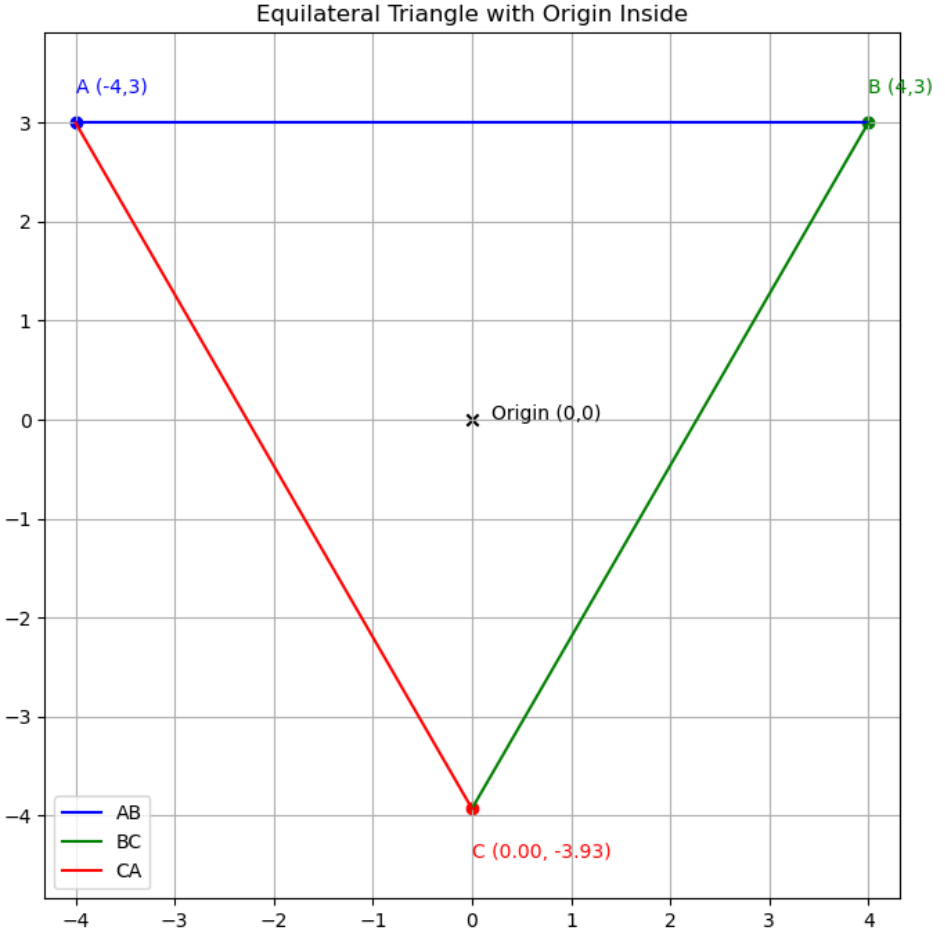


Fig. 0.1